1. 随机过程是给定时间的随机变量

2. 求概率密度函数 $f(x;t)=f(\theta_1)|\theta_1'(x)|+f(\theta_2)|\theta_2'(x)| = \frac{f(\theta_1)}{|x'(\theta_1)|}+\frac{f(\theta_2)}{|x'(\theta_2)|}$

3. 平稳随机过程 $X(t)$ 是在时间平移下概率性质不变的随机过程

4. 自相关函数的性质：① $R_X(0)=E[X^2(t)]$ 平均功率 ② $R_X(\tau)=R_X(-\tau)$ $K_X(\tau)=K_X(-\tau)$ 偶函数

③ $R_X(0)\geq|R_X(\tau)|$ $K_X(0)\geq|K_X(\tau)|$. ④ 周期→自相关周期．相等 (⑤ 分量) ⑤ $R_X(\infty)=m_X^2$

⑥ $\sigma_X^2=R_X(0)-R_X(\infty)$

5. 联合宽平稳的互相关函数． $R_{XY}(\tau)=R_{YX}(-\tau)$ $|R_{XY}(\tau)|^2\leq R_X(0)R_Y(0)$ $|R_{XY}(\tau)|\leq\frac{1}{2}[R_X(0)+R_Y(0)]$

6. 复随机过程 $R_Z(t,t+\tau)=E[Z^*(t)Z(t+\tau)]$ $K_Z(t,t+\tau)=R_Z(t,t+\tau)-m_Z^*(t)m_Z(t+\tau)$ 共轭

7. 正态随机过程 $f_X(x_1,x_2,\dots x_n;t_1,t_2,\dots,t_n)=\frac{1}{(2\pi)^{\frac{n}{2}}|K|^{\frac{1}{2}}}\exp\left[-\frac{(X-m_X)^T K^{-1}(X-m_X)}{2}\right]$ $K=\begin{bmatrix}k_{11} & k_{12} & \dots & k_{1n}\\ k_{21} & k_{22} & \dots & k_{2n}\\ \vdots & & & \\ k_{n1} & k_{n2} & \dots & k_{nn}\end{bmatrix}$

$k_{ij}=k_X(t_i,t_j)=E[(x_i-m_X(t_i))(x_j-m_X(t_j))]$

二维 $f_X(x_1,x_2;\tau_1)=\frac{1}{2\pi\sigma_X^2\sqrt{1-r^2(\tau_1)}}\exp\left[-\frac{(x_1-m_X)^2-2r(\tau_1)(x_1-m_X)(x_2-m_X)+(x_2-m_X)^2}{2\sigma_X^2[1-r^2(\tau_1)]}\right]$

8. 离散时间随机过程 遍历性 $A\langle X(n)\rangle=\overline{X(n)}=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}X_n$

时间自相关函数 $\mathcal{R}_{X_n}(n,n+m)=\overline{X(n)X(n+m)}=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}x_n x_{n+m}$

联合遍历 $\mathcal{R}_{X_nY_n}(n,n+m)=\overline{X(n)Y(n+m)}=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}X(n)Y(n+m)$

9. 马尔可夫过程：无后效性

马氏链的绝对~~分布~~概率由初始分布及一步转移概率确定

f_{ij} 迟早到达概率　$f_{ij}(n)$ 首达概率　$f_{ij}(\infty)=1-f_{ij}$

P_{ij} 1步 $i\to j$　$P_{ij}(n)$ n 步 $i\to j$　$P_{ij}=P_{ij}(1)$

$\boxed{f_{jj}=1}\xrightarrow{Y}$ 常返 $\longrightarrow$ 平均返回时间 $\boxed{\mu_i<\infty}\xrightarrow{Y}$ 正常返

$f_{jj}=1$ $\downarrow N$ 非　$\mu_i<\infty$ $\downarrow N$ 零

由于该链状态有限．所以至少有一个状态是常返的．并且该链是遍历的、所以各状态之间互通，

因此该链所有状态都是常返的

10. 泊松过程 $\frac{(\lambda t)^k}{k!}e^{-\lambda t}$

$E[X(t_a)-X(t_b)]=\lambda(t_a-t_b)$　$D[X(t_a)-X(t_b)]=\lambda(t_a-t_b)$　$E[(X(t_a)-X(t_b))^2]=\lambda^2(t_a-t_b)^2+\lambda(t_a-t_b)$

11. 电报信号 P14

半随机电报信号 $E[X(t)]=e^{-2\lambda t}$　$R_X(t_1,t_2)=e^{-2\lambda|\tau|}$

随机电报信号 $E[Y(t)]=0$　$R_Y(\tau)=e^{-2\lambda|\tau|}$.

1. P89 常见傅里叶变换对

2. 互谱密度性质 $S_{XY}(w)=S_{YX}(-w)=S_{YX}^*(w)$ $S_{XY}(w)$ 与 $S_{YX}(w)$ 互为共轭

$Re[S(\ )]$ 为偶 $Im[\ ]$ 奇 $S_{XY}(w)=\int_{-\infty}^{+\infty}R_{XY}(\tau)\cos w\tau d\tau-j\int_{-\infty}^{+\infty}R_{XY}(\tau)\sin w\tau d\tau$

3. 离散时间功率谱 $S_X(w)=\sum_{m=-\infty}^{\infty}R_X(m)e^{-jwm} \longleftrightarrow R_X(m)=\frac{1}{2\pi}\int_{-\pi}^{\pi}S_X(w)e^{jwm}dw$

$S_X'(z)=\sum_{m=-\infty}^{\infty}R_X(m)z^{-m}$ $S_X(w)=S_X'(z)|_{z=e^{jw}}$

4. 平稳随机过程的采样定理 $X(t)=\lim_{N\to\infty}\sum_{n=-N}^{N}X(nT_s)\,Sa\left[\frac{\pi(t-nT_s)}{T_s}\right]$

功率谱 $S(w)=\frac{1}{T_s}\sum_{n=-\infty}^{\infty}S_c\left(\frac{w+2\pi n}{T_s}\right)$

5. 低通型带限白噪声的自相关函数 $R_X(\tau)=\frac{WS_0}{\pi}\frac{\sin W\tau}{W\tau}$

带 $R_X(\tau)=\frac{WS_0}{\pi}\frac{\sin\frac{W\tau}{2}}{\frac{W\tau}{2}}\cos w_0\tau$

1. 随机信号通过连续时间系统

时域 均值 $m_Y(t)=m_X(t)*h(t)$ 频域 $m_Y=m_XH(0)$

平稳 $m_Y=m_X\int_0^{\infty}h(\tau)d\tau$

互相关 $R_{XY}(\tau)=R_X(\tau)*h(\tau)$

$R_{YX}(\tau)=R_X(\tau)*h(-\tau)$

自相关 $R_Y(\tau)=R_X(\tau)*h_1(\tau)*h(-\tau)$

$=R_{XY}(\tau)*h(-\tau)=R_{YX}(\tau)*h(\tau)$

$S_{XY}(w)=S_X(w)H(w)$

$S_{YX}(w)=S_X(w)H(-w)$

$S_Y(w)=S_X(w)|H(w)|^2$

$=S_{XY}(w)H(-w)=S_{YX}(w)H(w)$

输入随机信号带宽远大于线性系统带宽，可把输入信号看作白噪声

$\int_0^{\infty}\frac{\sin^2 ax}{x^2}dx=\frac{\pi}{2}|a|$ $\int_0^{\infty}\frac{\sin ax}{x}dx=\frac{\pi}{2}$

2. 离散时间系统 $Y(n)=\sum_{k=0}^{\infty}h(k)X(n-k)$

时域 $m_Y(n)=\sum_{k=0}^{\infty}h(k)E[X(n-k)]$ 频域 $m_Y=m_XH(1)$

平：$m_Y=m_X\sum_{k=0}^{\infty}h(k)$

$R_{XY}(m)=\sum_{k=0}^{\infty}h(k)R_X(m-k)=h(m)*R_X(m)$

$R_{YX}(m)=\sum_{k=0}^{\infty}h(k)R_X(m+k)=h(-m)*R_X(m)$

$R_Y(m)=R_X(m)*h(m)*h(-m)$

$=R_{XY}(m)*h(-m)=R_{YX}(m)*h(m)$

$S_{XY}(z)=H(z)S_X(z)$ $S_{XY}(w)=H(e^{jw})S_X(w)$

$S_{YX}(z)=H(z^{-1})S_X(z)$ $S_{YX}(w)=H(e^{-jw})S_X(w)$

$S_Y(z)=S_X(z)H(z)H(z^{-1})=S_{XY}(z)H(z^{-1})=S_{YX}(z)H(z)$

$R_Y(m)=\frac{1}{2\pi j}\oint_c S_Y(z)z^{m-1}dz=\frac{1}{2\pi}\int_{-\pi}^{\pi}|H(e^{jw})|^2S_X(w)e^{jwm}dw$

$E[Y^2(n)]=\frac{1}{2\pi j}\oint_c H(z)H(z^{-1})S_X(z)z^{-1}dz$

$=\frac{1}{2\pi}\int_{-\pi}^{\pi}|H(e^{jw})|^2S_X(w)dw$

3. 等效噪声带宽原则

1. 理想系统与实际系统在同一白噪声输入功率的激励下，两个系统的输出的平均功率相等

2. 理想系统的增益等于实际系统的最大增益

4. 白噪声通过理想线性系统

· 低通 $|H(\omega)|=\begin{cases}A & |\omega|\le\frac{\Delta\omega}{2}\\0 & 其他\end{cases}$ ← $\omega_0=0$ · 带通 $|H(\omega)|=\begin{cases}A & |\omega\pm\omega_0|\le\frac{\Delta\omega}{2}\\0 & 其他\end{cases}$

$S_X(\omega)=\frac{N_0}{2}$ $\quad S_Y(\omega)=\begin{cases}\frac{1}{2}N_0A^2 & |\omega|\le\frac{\Delta\omega}{2}\\0 & 其他\end{cases}$

$R_Y(\tau)=\frac{1}{2\pi}\int_{-\infty}^{\infty}S_Y(\omega)e^{j\omega\tau}d\omega=\frac{N_0A^2\Delta\omega}{4\pi}\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}$

$E[Y^2(t)]=\frac{N_0A^2\Delta\omega}{4\pi}$

相关系数 $r_Y(\tau)=\frac{K_Y(\tau)}{K_Y(0)}=\frac{R_Y(\tau)}{R_Y(0)}=\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}$

相关时间 $\tau_0=\int_0^{\infty}r_Y(\tau)d\tau=\int_0^{\infty}\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}d\tau=\frac{2}{\Delta\omega}\int_0^{\infty}\frac{\sin\frac{\Delta\omega\tau}{2}}{\tau}d\tau$

$=\frac{\pi}{\Delta\omega}$

$R_Y(\tau)=\frac{1}{2\pi}\int_{-\infty}^{\infty}S_Y(\omega)e^{j\omega\tau}d\omega=\frac{A^2N_0\Delta\omega}{2\pi}\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}\cos\omega_0\tau$

$E[Y^2(t)]=\frac{N_0A^2\Delta\omega}{2\pi}$

$r_Y(\tau)=\frac{R_Y(\tau)}{R_Y(0)}=\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}\cos\omega_0\tau$

$\tau_0=\int_0^{\infty}\frac{\sin\frac{\Delta\omega\tau}{2}}{\frac{\Delta\omega\tau}{2}}d\tau=\frac{\pi}{\Delta\omega}$

5. 对$Y(t)$进行采样，要求相邻点之间统计独立：$\tau_0\ll T$ 例题 $f_s=\frac{1}{\tau}\rightarrow R_X(\tau)=0$时$\tau$

6. $E\left[\int_0^1\int_0^1 X(u)X(v)dudv\right]=\int_0^1\int_{-u}^{1-u}R_X(\tau)dud\tau$

7. 希尔伯特变换 $X(t)\underset{\hat{X}(t)*-\frac{1}{\pi t}}{\overset{X(t)*\frac{1}{\pi t}}{\rightleftarrows}}\hat{X}(t)$ $\quad H_H(j\omega)=-j\,\mathrm{sgn}(\omega)$ $\quad H_{H^{-1}}(j\omega)=j\,\mathrm{sgn}(\omega)$

8. 解析过程 $\tilde{X}(t)=X(t)+j\hat{X}(t)$

性质 $S_{\hat{X}}(\omega)=S_X(\omega)$ $\quad R_{\hat{X}}(\tau)=R_X(\tau)$

$R_{\hat{X}X}(\tau)=-\hat{R}_X(\tau)$ $\quad R_{X\hat{X}}(\tau)=\hat{R}_X(\tau)=-R_{\hat{X}X}(\tau)=R_{\hat{X}X}(-\tau)$

$R_{\hat{X}X}(0)=R_{X\hat{X}}(0)=0$

$R_{\tilde{X}}(\tau)=2[R_X(\tau)+jR_{X\hat{X}}(\tau)]=2[R_X(\tau)+j\hat{R}_X(\tau)]$

$S_{X\hat{X}}(\omega)=\begin{cases}-jS_X(\omega), & \omega\ge0\\ jS_X(\omega) & \omega<0\end{cases}$ $\quad S_{\tilde{X}}(\omega)=\begin{cases}4S_X(\omega), & \omega\ge0\\0 & \omega<0\end{cases}$

9. 窄带随机过程

莱斯表达式 $X(t)=a(t)\cos\omega_0t-b(t)\sin\omega_0t$

$a(t)=X(t)\cos\omega_0t+\hat{X}(t)\sin\omega_0t$

$b(t)=-X(t)\sin\omega_0t+\hat{X}(t)\cos\omega_0t$

准正弦振荡表达式 $X(t)=A(t)\cos[\omega_0t+\varphi(t)]$

$A(t)=\sqrt{a^2(t)+b^2(t)}$ $\quad \varphi(t)=\arctan\frac{b(t)}{a(t)}$

性质: $E[a(t)]=E[b(t)]=0$ $\quad E[a^2(t)]=E[b^2(t)]=E[X^2(t)]$

$R_a(\tau)=R_b(\tau)=R_X(\tau)\cos\omega_0\tau+\overset{\hat{R}_X(\tau)}{R_{X\hat{X}}(\tau)}\sin\omega_0\tau$

$R_{ab}(\tau)=-R_X(\tau)\sin\omega_0\tau+R_{X\hat{X}}(\tau)\cos\omega_0\tau$ $\quad R_{ab}(0)=0$ $\quad R_{ab}(\tau)=-R_{ba}(\tau)=R_{ab}(-\tau)$

$R_X(\tau)=R_a(\tau)\cos\omega_0\tau+R_{ba}(\tau)\sin\omega_0\tau$

$S_a(\omega)=S_b(\omega)=\mathrm{lowpass}\,[S_X(\omega+\omega_0)+S_X(\omega-\omega_0)]$

$S_{ab}(\omega)=-j\,\mathrm{lowpass}\,[S_X(\omega+\omega_0)-S_X(\omega-\omega_0)]$ 若$S_X(\omega)$关于ω_0对称，$S_{ab}(\omega)=0$

10. 窄带过程包络与相位

$$f_{ab}(a_t,b_t)=\frac{1}{2\pi\sigma^2}e^{-\frac{a_t^2+b_t^2}{2\sigma^2}}$$

$\downarrow$ ①

$$f_{A\varphi}(A_t,\varphi_t)=\frac{A_t}{2\pi\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}\quad A_t\geq 0,\ 0\leq\varphi\leq 2\pi$$

↑ 同一时刻 A_t、φ_t 独立

$A(t)$、$\varphi(t)$ 独立 ✗

$\downarrow$ ②

$$f_A(A_t)=\frac{A_t}{\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}\qquad f_\varphi(\varphi_t)=\frac{1}{2\pi}$$

$\downarrow$ ③

$$f_A(A_t^2)=\frac{1}{2\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}$$

① $f_{A\varphi}(A_t,\varphi_t)=|J|\,f_{ab}(a_t,b_t)$

$$J=\begin{vmatrix}\frac{\partial h_1}{\partial A_t} & \frac{\partial h_1}{\partial\varphi_t}\\ \frac{\partial h_2}{\partial A_t} & \frac{\partial h_2}{\partial\varphi_t}\end{vmatrix}=\begin{vmatrix}\cos\varphi_t & -A_t\sin\varphi_t\\ \sin\varphi_t & A_t\cos\varphi_t\end{vmatrix}=A_t$$

② $f_A(A_t)=\int_0^{2\pi}f_{A\varphi}(A_t,\varphi_t)\,d\varphi_t$

$f_\varphi(\varphi_t)=\int_0^{\infty}f_{A\varphi}(A_t,\varphi_t)\,dA_t$

③ 令 $u_t=A_t^2$

$$f(u_t)=f_A(A_t)\,|h'(u_t)|=\frac{\sqrt{u_t}}{\sigma^2}e^{-\frac{u_t}{2\sigma^2}}\cdot\frac{1}{2\sqrt{u_t}}=\frac{1}{2\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}$$

1. 平方律检波器 (1) $y=bx^2$

· 正态噪声

$$E[X^n]=\begin{cases}1\times3\times5\times\cdots\times(n-1)\sigma^n, & n\geq2\text{ 的偶数}\\ 0 & \text{—}\end{cases}\qquad E[Y^n]=b^nE[X^{2n}]=b^n\,1\times3\times5\times\cdots\times(2n-1)\sigma_x^{2n}$$

$$\left.\begin{aligned}E[y(t)]&=b\sigma_x^2\\ E[y^2(t)]&=3b^2\sigma_x^4\\ &=3(E[y(t)])^2\end{aligned}\right\}\qquad\begin{aligned}\sigma_{y(t)}^2&=E[y^2(t)]-(E[y(t)])^2=2b^2\sigma_x^4=2(E[y(t)])^2\\ R_Y(\tau)&=b^2\sigma_x^4+2b^2R_x^2(\tau)\end{aligned}$$

· 信号与噪声同时：$x(t)=s(t)+n(t)$ （$s(t)$ 与 $n(t)$ 不相关） $s(t)=a\cos(\omega_0t+\theta)$

$$E[y(t)]=bE[\,s(t)]^2+\underset{0}{\underset{\|}{2s(t)n(t)}}+n(t)]^2\,]=b\left(\frac{a^2}{2}+\sigma_n^2\right)$$

$$R_Y(\tau)=b^2E[(s(t_1)+n(t_1))^2(s(t_2)+n(t_2))^2]=b^2E[s^2(t_1)s^2(t_2)+4s(t_1)s(t_2)n(t_1)n(t_2)+s^2(t_1)n^2(t_2)+s^2(t_2)n^2(t_1)+n^2(t_1)n^2(t_2)]$$

$$=b^2[R_{s^2}(\tau)+4R_s(\tau)R_n(\tau)+2\sigma_s^2\sigma_n^2+R_{n^2}(\tau)]=b^2\left(\frac{a^2}{2}+\sigma_n^2\right)^2+2b^2R_n^2(\tau)+2b^2a^2R_n(\tau)\cos\omega_0\tau+\frac{a^4b^2}{8}\cos2\omega_0\tau$$

2. 线性半波检波器 (1) $y=\begin{cases}bx & x>0\\ 0 & x\leq0\end{cases}$

输入 $f_X(x)=\frac{1}{\sqrt{2\pi}\sigma_x}e^{-\frac{x^2}{2\sigma_x^2}}$

$$E[y(t)]=b\int_0^{\infty}\frac{x}{\sqrt{2\pi}\sigma_x}e^{-\frac{x^2}{2\sigma_x^2}}=-\frac{b\sigma_x}{\sqrt{2\pi}}\int_0^{\infty}e^{-\frac{x^2}{2\sigma_x^2}}d\left(-\frac{x^2}{2\sigma_x^2}\right)=\frac{b\sigma_x}{\sqrt{2\pi}}$$

$$E[y^2(t)]=b^2E[x^2(t)]=b^2\int_0^{\infty}\frac{x^2}{\sqrt{2\pi}\sigma_x}e^{-\frac{x^2}{2\sigma_x^2}}dx=-\frac{b^2\sigma_x^2}{\sqrt{2\pi}\sigma_x}\int_0^{\infty}x\,de^{-\frac{x^2}{2\sigma_x^2}}=\frac{b^2\sigma_x}{\sqrt{2\pi}}\int_0^{\infty}e^{-\frac{x^2}{2\sigma_x^2}}dx=\frac{1}{2}b^2\sigma_x^2$$

$$\sigma_y^2=E[y^2(t)]-(E[y(t)])^2=\frac{b^2\sigma_x^2}{2}-\frac{b^2\sigma_x^2}{2\pi}=\frac{b^2\sigma_x^2}{2}\left(1-\frac{1}{\pi}\right)$$

3. 特征函数

$$R_Y(\tau)=E[y(t)y(t+\tau)]=E\left[\frac{1}{2\pi}\int_{-\infty}^{\infty}F(u)e^{jux(t)}du\cdot\frac{1}{2\pi}\int_{-\infty}^{\infty}F(v)e^{jvx(t+\tau)}dv\right]=\frac{1}{4\pi^2}\int_{-\infty}^{\infty}F(u)du\int_{-\infty}^{\infty}F(v)\underbrace{E[e^{jux(t)+jvx(t+\tau)}]}_{M_2(u,v;\tau)}dv$$

$$E[e^{jux(t)+jvx(t+\tau)}]=M_2(u,v;\tau)\longrightarrow R_Y(\tau)=\frac{1}{4\pi^2}\int_{-\infty}^{\infty}F(u)du\int_{-\infty}^{\infty}F(v)M_2(u,v;\tau)dv$$

$$=\frac{1}{(2\pi j)^2}\int_D F(s_1)\int_D F(s_2)M_2(u,v;\tau)ds_1ds_2$$

例：输入 $E[x(t)]=0$，$E[x^2(t)]=1$ 的正态噪声 $M_2(s_1,s_2,\tau)=e^{\frac{1}{2}[s_1^2+s_2^2+2s_1s_2R_x(\tau)]}$ $\quad e^{s_1s_2R_x(\tau)}=\frac{[s_1s_2R_x(\tau)]^k}{k!}$

4. 准正弦振荡信号

输入：$x(t)=A(t)\cos[\omega_0 t+\varphi(t)]$ $\longrightarrow$ 输出 $y(t)=f[A_t\cos(\omega_0 t+\varphi_t)]=f_0(A_t)+f_1(A_t)\cos\phi_t+\cdots$

统计特性：均 $E[y_t]=E[f_0(A_t)]=\int_0^{\infty}f_0(A_t)f_A(A_t)dA_t$ 低频分量：$\frac{1}{\pi}\int_0^{\pi}f(A_t\cos\phi_t)d\phi_t$

$\sigma^2=E[I_0^2]-(E[I_0])^2=\int_0^{\infty}f_0^2(A_t)f_A(A_t)dA_t-[\int_0^{\infty}f_A(A_t)f_A(A_t)dA_t]^2$

5. 线性半检波器(b2)

输入：$x_t=A_t\cos\phi_t$ $\longrightarrow$ 输出：$y_t=f(x_t)=\begin{cases}bA_t\cos\phi_t & -\frac{\pi}{2}<\phi_t<\frac{\pi}{2}\\ 0 & 其他\end{cases}$ $\xrightarrow{低频}$ $I_0=\frac{bA_t}{\pi}$

$E[I_0]=\int_0^{\infty}I_0 f_{A_t}(A_t)dA_t=\int_0^{\infty}\frac{bA_t}{\pi}\frac{A_t}{\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}dA_t=\frac{b\sigma}{\sqrt{2\pi}}$

$\sigma^2=E[I_0^2]-(E[I_0])^2=\int_0^{\infty}I_0^2 f_{A_t}(A_t)dA_t-\frac{b^2\sigma^2}{2\pi}=\frac{b^2\sigma^2}{2\pi^2}(4-\pi)$

6. 平方律检波器(2) $y=bx^2$

$E[y_t]=E[bA_t^2\cos^2\phi_t]=bE[A_t^2]E[\cos^2\phi_t]=b\sigma^2$

$y_t=b(A_t\cos\phi_t)^2=\frac{1}{2}bA_t^2(1+\cos 2\phi_t)$ $\xrightarrow{低频}$ $\frac{1}{2}bA_t^2$

$\sigma^2=E[y_t^2]-(E[y_t])^2=\int_0^{\infty}y_t^2 f_{A_t}(A_t)dA_t-b^2\sigma^4=\int_0^{\infty}(\frac{1}{2}bA_t^2)^2\frac{A_t}{\sigma^2}e^{-\frac{A_t^2}{2\sigma^2}}dA_t-b^2\sigma^4=2b^2\sigma^4-b^2\sigma^4=b^2\sigma^4$

1. 无偏估计：待估参数 a 和它的估计值 $\hat{a}$ 的均值 $E(\hat{a})$ 相等

估计质量：偏差和方差结合起来.

一致估计：随着样本容量的增加、估计的均方误差趋于零

2. 数字特征的估计

$\hat{m}_{x(N)}=\frac{1}{N}\sum_{n=0}^{N-1}x_n$

$\hat{\sigma}^2_{x(N)}=\frac{1}{N-1}\sum_{n=0}^{N-1}(x_n-m_{x(N)})^2$ m_x 未知 $\hat{\sigma}^2_{x(N)}=\frac{1}{N-1}\sum_{n=0}^{N-1}(x_n-\hat{m}_{x(N)})^2$

$\hat{R}_x(m)=\frac{1}{N-|m|}\sum_{n=0}^{N-|m|-1}x_n x_{n+m}$ $m=0,\pm1,\pm2\ldots$

3. 最大似然估计(分布函数或概率密度未知)

$p(x;\theta)=p(x_1,x_2,\ldots,x_N;\theta)=\prod_{k=1}^{N}p(x_k;\theta)$

$L(\theta)=\ln\prod_{k=1}^{N}p(x_k;\theta)$

令 $\frac{\partial L(\theta)}{\partial\theta}=0$ 解得 $\hat{\theta}=$

$N\to\infty$ 时，最大似然无偏、一致、有效、正态分布.